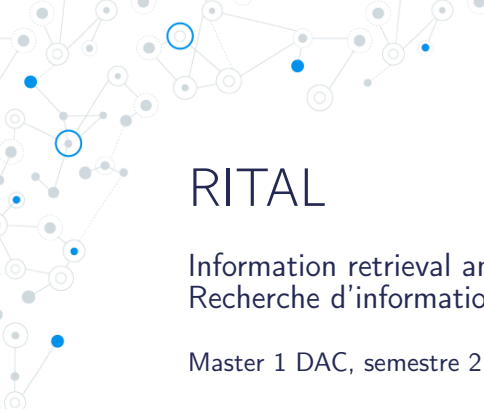




RITAL

Information retrieval and natural language processing
Recherche d'information et traitement automatique de la langue

Master 1 DAC, semestre 2

Laure Soulier - basés sur les slides de V. Guigue et N. Thome



- 1 Organisation de l'UE
- 2 A guided tour on NLP applications
- 3 Main methods and evaluation framework in NLP
- 4 Example of task: document classification (Project)

Organisation de l'UE

Les données textuelles:

- correspondent à de nombreuses applications et **débouchés professionnels**
- nécessitent des **outils spécifiques...**
- ... qui ont beaucoup évolué ces dernières années
- en plus des modèles, le **savoir faire** et les **campagnes d'expériences** sont très importants
- une **évaluation** souvent délicate

⇒ RITAL = vous donner les clés pour attaquer ces problèmes

1 TAL: Traitement Automatique de la Langues

- Analyse du texte, classement de documents, compréhension des phrases, etc...
 - Séances 1-5 : Laure Soulier (CM), Salim Aissi (TME - DAC), Matthieu Dubois et Zofia MILCZAREK (TME - ISI)

2 RI: Recherche d'Information - pour le parcours DAC

- Stockage, indexation, accès
 - Séances 5-10 : Laure Soulier (CM) et Salim Aissi (TME)

Mots clés

Représentation de textes, modèles, campagnes d'expériences, savoir-faire opérationnel

- Canal mattermost : <https://channel.isir.upmc.fr/m1-mind-25-27/channels/rital>

25% **Projet TAL**

- Notebooks sur la classification supervisée, non supervisée, introduction à l'apprentissage de représentations
- Performances en classification de document (x2)
- Rapport sur les campagnes d'expériences

25% **Projet RI**

- Ré-implémentation d'un article scientifique
- Expérimentations
- Présentation orale

50% **Examen final**

- Mises en situation
- Formulations légères

Traitement Automatique de la Langue Naturelle = Natural Language Processing

Divisé en de multiples applications à différents niveaux d'analyse

Au moins une source à étudier: C. Manning, Stanford:

<https://nlp.stanford.edu/cmanning/>

<http://web.stanford.edu/~jurafsky/NLPCourseraSlides.html>

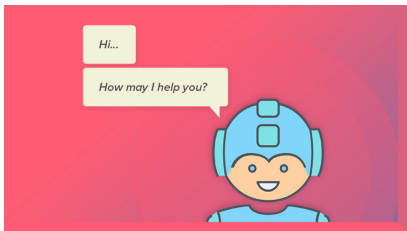
<http://web.stanford.edu/class/cs224n/>

A guided tour on NLP applications

- 1 Organisation de l'UE
- 2 A guided tour on NLP applications**
 - Context
 - NLP tasks
- 3 Main methods and evaluation framework in NLP
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Why Natural Language Processing?

- Access Knowledge (search engine, recommender system...)
- Communicate (e.g. Translation)
- Linguistics and Cognitive Sciences (Analyse Languages themselves)



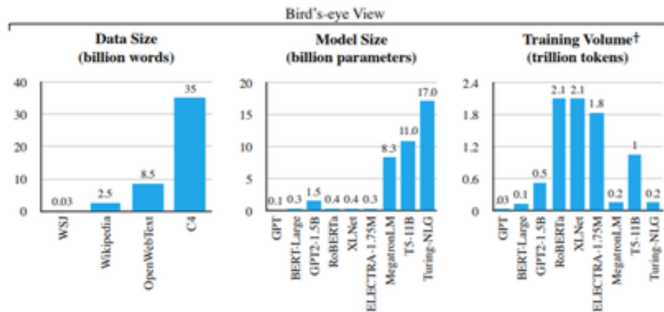
I		WANT		FORTY		KILOGRAMS OF		PERSIMMONS
↓		↓		↓		↓		↓
ICH		WOLLEN		VIERZIG		KILOGRAMM		PERSIMONEN

- Natural language (french, english, ...) , constructed language (esperanto) - continuum or clear difference?
- Structure: letters, sounds, words, narrative, ...
- Phonemes (pronunciation) / Morphèmes (meaning)
→ **How to represent the language? How does the language work?**
- Communication mean (oral, writen, signed, programming language! ...) vs (stone, papyrus, paper, typewriters, computers, phones, ...)

- 70 billion web-pages online (1.9 billion websites)
- 55 million Wikipedia articles
- 9000 tweets/second
- 3 million mail / second (60% spam)

**In NLP,
Everything is Big
and Getting
Bigger**

credit: AI21 Labs



Potential Users of NLP

- 7.9 billion people use some sort of language (January 2022)
- 4.7 billion internet users (January 2021) (59%)
- 4.2 billion social media users (January 2021) (54%)

Products

- Search: +2 billion Google users, 700 millions Baidu users
- Social Media: +3 billion users of Social media (Facebook, Instagram, WeChat, Twitter...)
- Voice assistant: +100 million users (Alexa, Siri, Google Assistant)
- Machine Translation: 500M users for google translate
- ChatGPT & cie : 300 millions d'utilisateurs hebdomadaires (dec 2024)

1 Productivity

- New words, senses, structure

- Ex: staycation (stay - vacation), social distance added to the Oxford Dictionary in '21

2 Ambiguity

3 Variability

4 Diversity

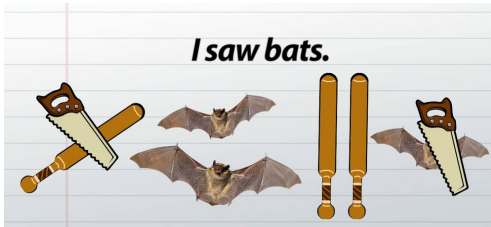
5 Sparsity

2. Ambiguity

- Having more than one meaning
- Disambiguation $\Rightarrow$ context, external knowledge

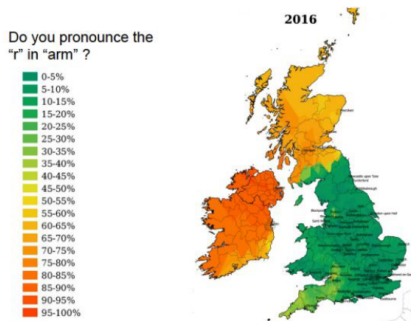
Type of ambiguities

- Semantic / lexical ambiguities: several possible meanings within a single word
 - Polysemy, e.g. set , arm, head: Head of New-Zeland is a woman
 - Name Entity: e.g. Michael Jordan (professor at Berkeley, basketball player)
 - Object/Color e.g. cherry (our cherry coat)
- Syntactic ambiguities: structural/grammatical ambiguity with a sentence



Variations at several levels: phonetic, syntactic, semantic

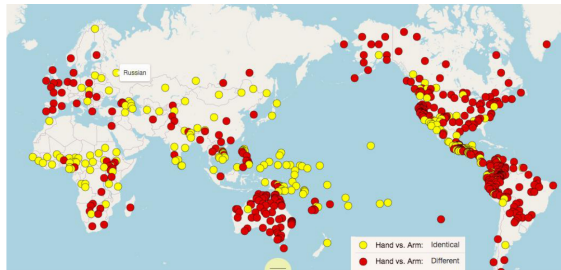
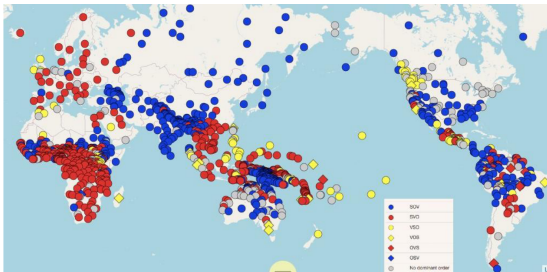
- Semantic: language variability wrt social context, geography, sociology, date, topic



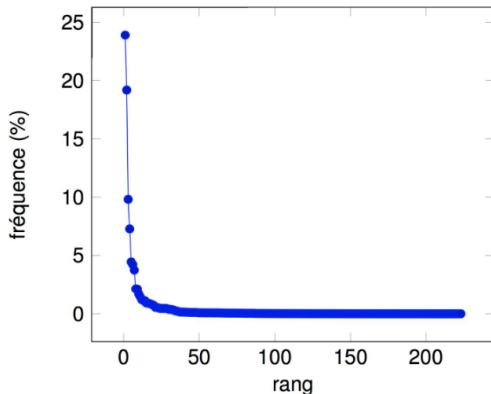
T'as vu il l'a bien cherché wsh #AperoChezRicard
> +10000, shah!
> tabuz, lavé rien fé
> ki ca ? le mec ou son chien ?
> Wtf is wrong with him ? #PETA4EVER
> ki ca ? le chien ?
> loool

BING translation:
You saw coming it #AperoChezRicard wsh
> +10000, shah!
> tabuz, washed anything fe
> Ki ca? the guy or his dog?
> WTF is wrong with him?
#PETA4EVER
> Ki ca? the dog?
> loool

- Syntactic, e.g. Subject (S) Verb (V) Object (O) (VOS) order
- Semantic, e.g. Words partition: hands vs arm



- Word's frequency $\sim$ Zipf law, *i.e.* k^{st} most frequent term: $f_w(k) \propto \frac{1}{k^\theta}$



- Issue: informative words/info in this distribution? $\Rightarrow$ more details soon



Why else is natural language understanding difficult?

non-standard English

Great job @justinbieber! Were SOO PROUD of what youve accomplished! U taught us 2 #neversaynever & you yourself should never give up either♥

segmentation issues

the New York-New Haven Railroad
the New York-New Haven Railroad

idioms

dark horse
get cold feet
lose face
throw in the towel

neologisms

unfriend
Retweet
bromance

world knowledge

Mary and Sue are sisters.
Mary and Sue are mothers.

tricky entity names

Where is *A Bug's Life* playing ...
Let It Be was recorded ...
... a mutation on the *for* gene ...

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- 1 At the document level
 - Topic/sentiment classification
 - Question-answering
 - Document ranking (information retrieval)
 - etc...
- 2 At the paragraph / sentence level
 - Co-reference resolution
- 3 At the word level
 - Synonymy
- 4 At the stream level
 - Topic detection & tracking
 - Dialogue management

⇒ Different levels often refer to different format of data (tabular, sequence, stream)

■ Indexing

- cf Information Retrieval



- **Indexing**
 - cf Information Retrieval
- **Classification / filtering**
 - topic



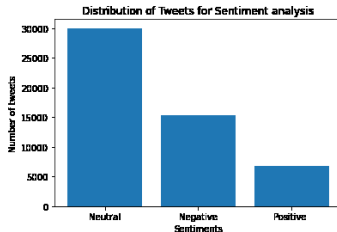
-



- Indexing
 - cf Information Retrieval
- Classification / filtering
 - topic
- Counting / survey
 - Sentiment classification

POSITIVE
Great product

NEGATIVE
Worst experience in my life



- **Indexing**
 - cf Information Retrieval
- **Classification / filtering**
 - topic
- **Counting / survey**
 - Sentiment classification
- **Clustering (topic analysis)**
 - Unsupervised formulation
 - Large corpus primary exploration
 - Lexical field extraction

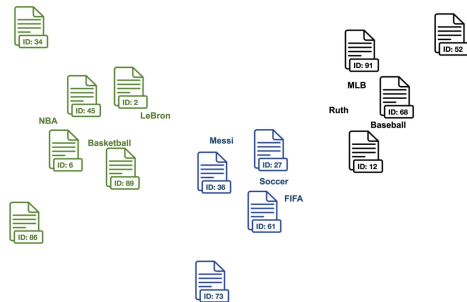
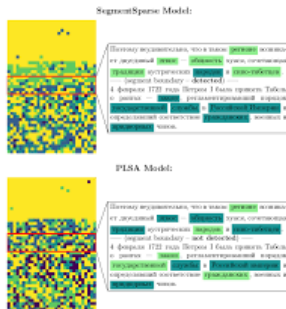


Figure 1: source: <https://towardsdatascience.com/unsupervised-text-classification-with-lbl2vec-6c5e040354de>

- Sentence classification
- Link with stream
- Break detection

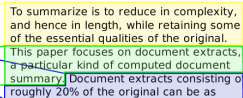


Тема 3
задание
9/2024
слова
слово



- Sentence extraction
- Generative architecture

Extractive Summary



■ Document Segmentation

- Sentence classification
- Link with stream
- Break detection

■ Automated summary

- Sentence extraction
- Generative architecture

■ Translation

- Mostly at the sentence level
- Mais cela change...

■ Question Answering (QA)

- Classification formulation
- Extraction approach (Siri/Google assistant)
- An emerging task... For several other tasks!

Passage Sentence

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under gravity.

Question

What causes precipitation to fall?

Answer Candidate

gravity

- Between **question** and **answer**

cause---gravity

precipitation---gravity

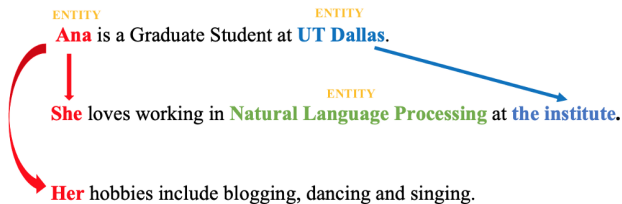
fall---gravity

what---gravity

■ Question Answering (QA)

- Classification formulation
- Extraction approach (Siri/Google assistant)
- An emerging task... For several other tasks!

■ Coreference resolution



■ Question Answering (QA)

- Classification formulation
- Extraction approach (Siri/Google assistant)
- An emerging task... For several other tasks!

■ Coreference resolution

■ Information extraction

- Mainly a sentence level task...
- Sometimes at the paragraph level (with coreference / QA)

Sebastian Thrun PERSON started at Google ORG in 2007 DATE, feeling him seriously. "I can tell you very senior CEOs of major American NORP and turn away because I wasn't worth talking to," said Thrun PERSON, the higher education startup Udacity, in an interview with Recode ORG.

less than a decade later DATE, dozens of self-driving startups have and the world clamor, wallet in hand, to secure their place in the fast-moving transportation.

■ Question Answering (QA)

- Classification formulation
- Extraction approach (Siri/Google assistant)
- An emerging task... For several other tasks!

■ Coreference resolution

■ Information extraction

- Mainly a sentence level task...
- Sometimes at the paragraph level (with coreference / QA)

■ Dialog State Tracking

- Chatbot...

Sam walks into the kitchen.
Sam picks up an apple.
Sam walks into the bedroom.
Sam drops the apple.

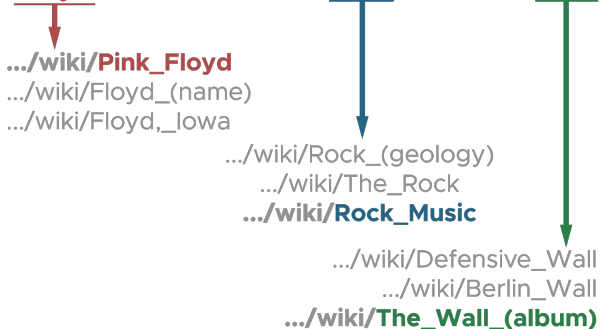
Q: Where is the apple?

A. Bedroom

State that for each question...

- Word/entity disambiguation
- Part-Of-Speech tagging (POS)
- Chunking (sentence segmentation)
- Semantic Role Labeling (SRL)
- Sentiment Analysis (fine grained)
- Named Entity Recognition (NER)
- Information Extraction
- Machine Translation

“Floyd revolutionized rock with the Wall”

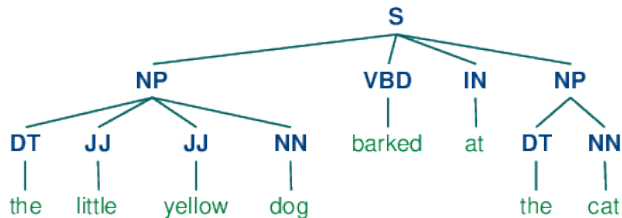


Word/entity disambiguation

- Word/entity disambiguation
- Part-Of-Speech tagging (POS)
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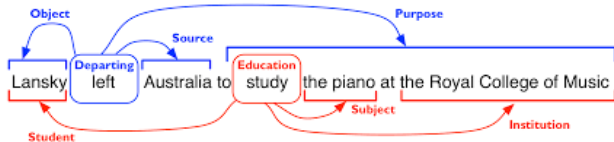


- Word/entity disambiguation
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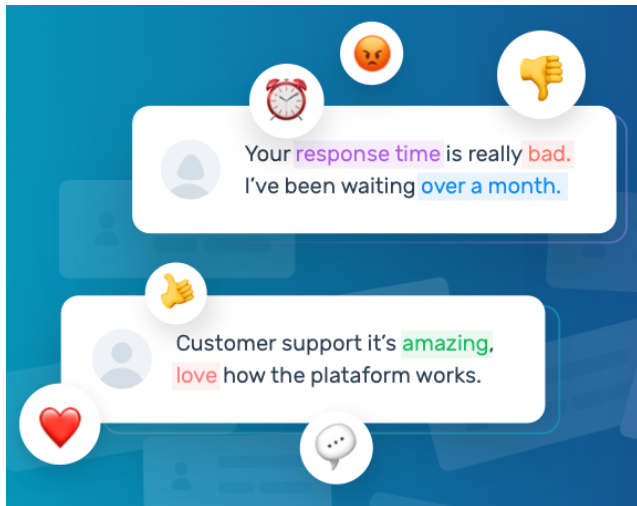
Chunking: Syntactic Parsing

- Word/entity disambiguation
- Part-Of-Speech tagging (POS)
- Chunking (sentence segmentation)
- Semantic Role Labeling (SRL)
- Sentiment Analysis (fine grained)
- Named Entity Recognition (NER)
- Information Extraction
- Machine Translation



SRL

- Word/entity disambiguation
- Part-Of-Speech tagging (POS)
- Chunking (sentence segmentation)
- Semantic Role Labeling (SRL)
- Sentiment Analysis (fine grained)
- Named Entity Recognition (NER)
- Information Extraction
- Machine Translation



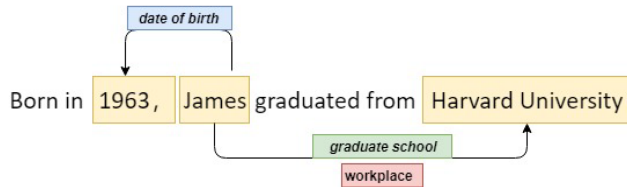
Fine-grained (word-level) sentiment analysis

- Word/entity disambiguation
- Part-Of-Speech tagging (POS)
- Chunking (sentence segmentation)
- Semantic Role Labeling (SRL)
- Sentiment Analysis (fine grained)
- Named Entity Recognition (NER)
- Information Extraction
- Machine Translation

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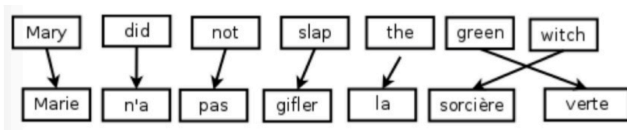
NER

- Word/entity disambiguation
- Part-Of-Speech tagging (POS)
- Chunking (sentence segmentation)
- Semantic Role Labeling (SRL)
- Sentiment Analysis (fine grained)
- Named Entity Recognition (NER)
- Information Extraction
- Machine Translation



Relation Extraction

- Word/entity disambiguation
- Part-Of-Speech tagging (POS)
- Chunking (sentence segmentation)
- Semantic Role Labeling (SRL)
- Sentiment Analysis (fine grained)
- Named Entity Recognition (NER)
- Information Extraction
- Machine Translation



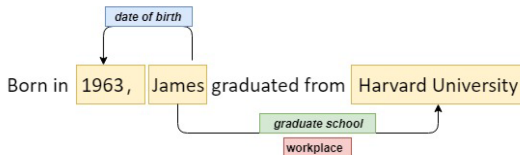
Translation

Text = lot of resources $\Rightarrow$ Universal knowledge?

- Noisy (both at the syntactic & semantic levels)
- Hard to query & exploit (search, knowledge inference,...)

(RDF) knowledge

- Def: Entity 1 (subject) Relation (predicate) Entity 2 (target)
- Simpler version: Key / value
- Easy to handle



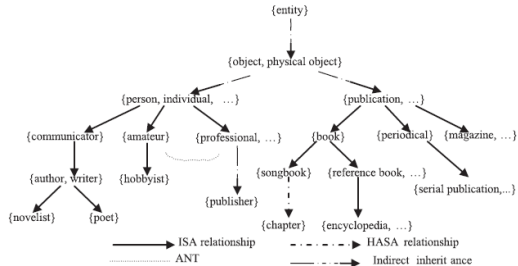
- Syntactic similarities
 - spelling correction
 - Levenshtein = DTW
- Semantic distance (synonymy)
 - WordNet (& other resources)
 - Representation learning

```
e}  
Correction "distnce": suggestions [fr]: distance, distancé  
Calcul de Aperçu du problème (⌘F8) Aucune solution disponible dar  
Calcul de distnce Sémantique  
n{itemize}  
  \item Ressources WordNet  
itemize}  
Dictionnaires en tous genres
```

- Syntactic similarities
 - spelling correction
 - Levenshtein = DTW
- Semantic distance (synonymy)
 - WordNet (& other resources)
 - Representation learning

		H	Y	U	N	D	A	I
	0	1	2	3	4	5	6	7
H	1	0	1	2	3	4	5	6
O	2	1	1	2	3	4	5	6
N	3	2	2	2	2	3	4	5
D	4	3	3	3	3	2	3	4
A	5	4	4	4	4	3	2	3

- Syntactic similarities
 - spelling correction
 - Levenshtein = DTW
- Semantic distance (synonymy)
 - WordNet (& other resources)
 - Representation learning



- Historical tasks: **Topic Detection and Tracking (TDT)**
- = adding a temporal axis to the topic detection
 - topic quantification
 - topic appearance
 - topic vanishing

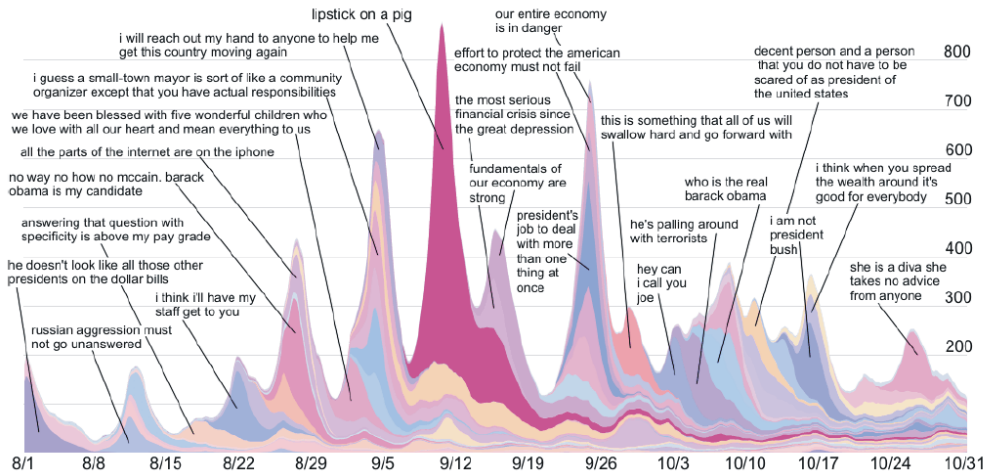


Figure 4: Top 50 threads in the news cycle with highest volume for the period Aug. 1 – Oct. 31, 2008. Each thread consists of all news articles and blog posts containing a textual variant of a particular quoted phrases. (Phrase variants for the two largest threads in each week are shown as labels pointing to the corresponding thread.) The data is drawn as a stacked plot in which the thickness of the strand corresponding to each thread indicates its volume over time. Interactive visualization is available at <http://memetracker.org>.

Profiling = recommender system

- modern User Interface (UI)
- Information Access
 - Personalization
 - Active suggestion (user = query)

Recent proposal: mixing user interaction & textual review

- user interaction = best item similarity
- review = in depth textual description

McAuley & Leskovec 2013



Language Technology

making good progress

mostly solved

Spam detection

Let's go to Agra!



Buy V1AGRA ...



Part-of-speech (POS) tagging

ADJ ADJ NOUN VERB ADV

Colorless green ideas sleep furiously.

Named entity recognition (NER)

PERSON ORG LOC

Einstein met with UN officials in Princeton

Sentiment analysis

Best roast chicken in San Francisco!



The waiter ignored us for 20 minutes.



Coreference resolution

Carter told Mubarak he shouldn't run again.

Word sense disambiguation (WSD)

I need new batteries for my *mouse*.



Parsing

I can see Alcatraz from the window!

Machine translation (MT)

第13届上海国际电影节开幕...



The 13th Shanghai International Film Festival...

Information extraction (IE)

You're invited to our dinner party, Friday May 27 at 8:30



Party
May 27
add

still really hard

Question answering (QA)

Q. How effective is ibuprofen in reducing fever in patients with acute febrile illness?

Paraphrase

XYZ acquired ABC yesterday

ABC has been taken over by XYZ

Summarization

The Dow Jones is up

The S&P500 jumped

Housing prices rose



Economy is good

Dialog

Where is Citizen Kane playing in SF?



Castro Theatre at 7:30. Do you want a ticket?



Main methods and evaluation
framework in NLP

- 1 Organisation de l'UE
- 2 A guided tour on NLP applications
- 3 Main methods and evaluation framework in NLP**
 - Main method families
 - Standard processing pipelines
 - Evaluation framework
- 4 Example of task: document classification (Project)

- Language theory: a language to define the language (e.g. regular expressions)
- Automata: inputs, states and transitions.

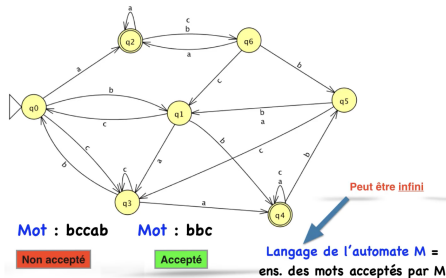


Figure 2: Source: https://www.youtube.com/watch?v=jAUQfvAkHrA&ab_channel=%C3%80lad%C3%A9couvertedesgraphes

- Pattern finding

Think as a data-scientist:

- No magic !
- For each application :
 - 1 Identify the problem class (regression, classification, ...)
 - 2 Find global Input / output
 - 3 Think about the data format (I/O)
 - 4 Find a model to deal with this data
 - 5 **Optimize all parameters & hyper-parameters**
 - Conduct an experiment campaign

$$\{(x_i, y_i)\}_{i=1, \dots, N} \quad \Rightarrow \quad \text{learn:} \quad f_{\theta, \phi}(x) \approx y$$

Let's have a tour on NLP applications !

1. Preprocessing

- encoding (latin, utf8, ...)
- punctuation
- stemming
- lemmatization
- tokenization
- capitals/lower case
- regex
- ...

2. Formatting

- Dictionary
- + reversed Index
- Vectorial format
- Sequence format

3. Learning

- Doc / sentence / word classification
- Linear/non-linear classifiers?
- HMM, CRF?
- Deep, transformers?

4. Hyper-parameter optimization

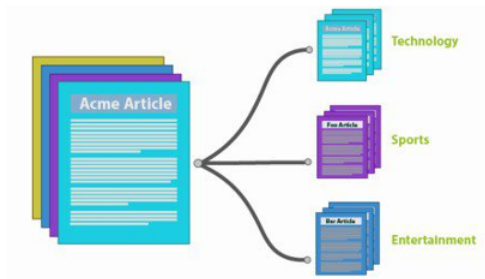
What is a good evaluation?

- Tasks and evaluation objectives well-defined
- Good datasets or possibility generating/annotating datasets
- Good metrics : accuracy is not the only metric! Depend on the task
- Good evaluation protocol: intrinsic and extrinsic evaluations
- Good training procedures: train/test split, cross-validation, ...
- Plenty of criterias: interpretability, environmental cost
(<https://calculator.green-algorithms.org/>, Carbon Tracker),

Example of task: document classification (Project)

- 1 Organisation de l'UE
- 2 A guided tour on NLP applications
- 3 Main methods and evaluation framework in NLP
- 4 Example of task: document classification (Project)**
 - Task definition
 - Main approach
 - Evaluation/outils

- Assigning a document (or paragraph) to a set of predefined categories: sport, science, medicine, religion, etc
- Classification task



2 tasks in project (practicals):

- Sentiment classification on movie reviews
- Speaker identification (Chirac/Mitterrand)

- Classification **thématique**
 - classer les news sur un portail d'information,
 - trier des documents pour la veille sur internet,
 - présenter les résultats d'une requête
- Classification **d'auteurs**
 - review spam,
 - détection d'auteurs
- Information pertinente/non pertinente
 - filtrage personnalisé (à partir d'exemple), classifieur actif (évoluant au fil du temps)
 - spam/non spam
- Classification de **sentiments**
 - documents positifs/négatifs, sondages en ligne

Données d'apprentissage:

```
<100:1:C> Quand je dis chers amis, ...  
<100:2:C> D'abord merci de cet ...  
...  
<100:14:M> Et ce sentiment ...
```

Le format est le suivant: <ID-Discours:ID-phrased:Etiquette>, C → Chirac, M → Mitterrand

Données de test, sans les étiquettes:

```
<100:1> Quand je dis chers amis, ...  
<100:2> D'abord merci de cet ...  
...
```


The Task

Building a model for movies revisions in English for classifying it into positive or negative.



Sentiment Polarity Dataset Version 2.0

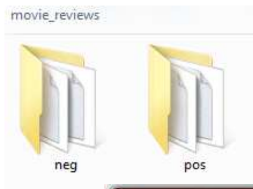
1000 positive movie review and 1000 negative review texts from:

Thumbs up? Sentiment Classification using Machine Learning Techniques. Bo Pang, Lillian Lee, and Shivakumar Vaithyanathan.
Proceedings of EMNLP, pp. 79--86, 2002.

“Our data **source** was the **Internet Movie Database** (IMDb) archive of the rec.arts.movies.reviews newsgroup.³ We selected only reviews where the **author rating** was **expressed** either with stars or some **numerical value** (other conventions varied too widely to allow for automatic processing). Ratings were automatically extracted and converted into one of three categories: positive, negative, or neutral. For the work described in this paper, we concentrated **only** on discriminating between **positive** and **negative** sentiment.”

<http://www.cs.cornell.edu/people/pabo/movie-review-data/>

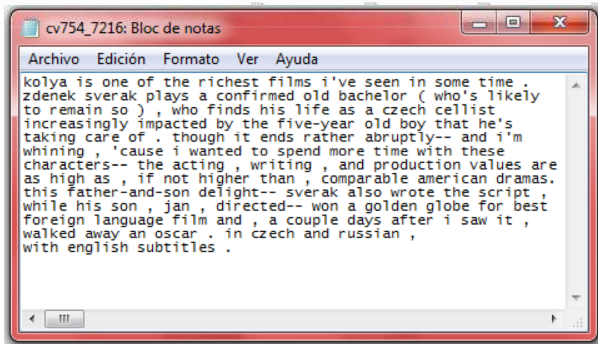
The Data (1/2)



Biblioteca Documentos

pos

cv754_7216	cv114_18398	cv938_10220	cv013_10159
cv280_8267	cv471_16858	cv424_8831	cv253_10077
cv825_5063	cv230_7428	cv763_14729	cv722_7110
cv057_7453	cv075_6500	cv319_14727	cv082_11080
cv640_5378	cv058_8025	cv430_17351	cv312_29377
			cv361_28944
			cv931_17563
			cv170_3006



Handling textual data: the classification case

- 1 Big corpus $\Leftrightarrow$ Huge vocabulary
- 2 Sentence structure is hard to model
- 3 Words are polymorphous: singular/plural, masculine/feminine
- 4 Synonyms: how to deal with?
- 5 Machine learning + large dimensionality = problems

Handling textual data: the classification case

- 1 Big corpus $\Leftrightarrow$ Huge vocabulary
Logistic Regression, SVM, Naive Bayes... Boosting, Bagging...
Distributed & efficient algorithms
- 2 Sentence structure is hard to model
Removing the structure...
- 3 Words are polymorphous: singular/plural, masculine/feminine
Several approaches... (wait for the next course !)
- 4 Synonyms: how to deal with?
wait for the next course !
- 5 Machine learning + large dimensionality = problems
Removing useless words

$\Rightarrow$ Pre-processing + Bag of Word (BoW) Model + Machine Learning

- 1 Organisation de l'UE
- 2 A guided tour on NLP applications
- 3 Main methods and evaluation framework in NLP
- 4 Example of task: document classification (Project)**
 - Task definition
 - **Main approach**
 - Evaluation/outils

1. Preprocessing

- encoding (latin, utf8, ...)
- punctuation
- stemming
- lemmatization
- tokenization
- capitals/lower case
- regex
- ...

2. BoW Model

- Dictionary
- Vectorial format
- TF-IDF
- Binary/non-binary
- N-grams

3. Learning

- Doc / sentence / paragraph classification
- Linear/non-linear classifiers?
- Naive Bayes, logistic regression, SVM

4. Hyper-parameter optimization

More details next course

- Très rapide, interprétable:
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 - ensemble des documents $\{d_i\}_{i=1,\dots,N}$,
 - documents = une suite de mots w_j : $d_i = (w_1, \dots, w_{|d_i|})$.
 - modèle Θ_c pour chaque classe de documents.
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- Modélisation naive: $P(d_i|\Theta_c) = \prod_{j=1}^{|d_i|} P(w_j|\Theta_c) = \prod_{j=1}^{|D|} P(w_j|\Theta_c)^{x_i^j}$
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 x_i^j décrit le nombre d'apparitions du mot j dans le document i
- Notation: $P(w_j|\Theta_c) \Rightarrow \Theta_c^j$,
 Résolution de: $\Theta_c = \arg \max_{\Theta} \sum_{i=1}^{|C|} \sum_{j=1}^{|D|} x_i^j \log \Theta_c^j$
 Solution: $\Theta_c^j = \frac{\sum_{d_i \in C} x_i^j}{\sum_{d_i \in C} \sum_{j \in d_i} x_i^j}$

- Très simple à calculer (possibilité de travailler directement en base de données)
- Naturellement multi-classes,

$$\text{inférence: } \arg \max_c \sum_{j=1}^{|D|} x_i^j \log(\Theta_c^j)$$

Performance intéressante... Mais améliorable

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- Extensions:

- Robustesse : $\Theta_c^j = \frac{\sum_{d_i \in C_m} x_i^j + \alpha}{\sum_{d_i \in C_m} \sum_{j \in d_i} x_i^j + \alpha |D|}$
- Mots fréquents (*stopwords*) ... Bcp d'importance dans la décision
- pas d'aspect discriminant

- Données en sacs de mots, différents codages possibles:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ \vdots & & & \\ x_{N1} & x_{N2} & \cdots & x_{Nd} \end{bmatrix}$$

Un document $x_i \in \mathbb{R}^d$

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- Décision linéaire:
 - Décision linéaire simple:

$$f(x_i) = x_i w = \sum_j x_{ij} w_j$$

- Régression logistique:

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- Mode de fonctionnement bi-classe (extension par un-contre-tous)

■ Formulation

- Maximisation de la vraisemblance ($y_i \in \{0, 1\}$):

$$L = \prod_{i=1}^N P(y_i = 1|x_i)^{y_i} \times [1 - P(y_i = 1|x_i)]^{1-y_i}$$

$$L_{\log} = \sum_{i=1}^N y_i \log(f(x_i)) + (1 - y_i) \log(1 - f(x_i))$$

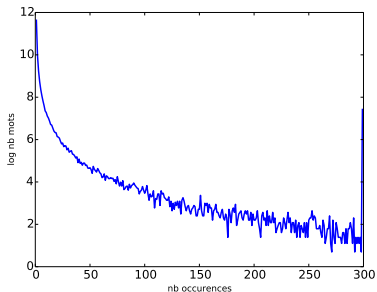
- Minimisation d'un coût ($y_i \in \{-1, 1\}$):

$$C = \sum_{i=1}^N (f(x_i) - y_i)^2 \quad \text{ou} \quad C = \sum_{i=1}^N (-y_i f(x_i))_+$$

- Passage à l'échelle: technique d'optimisation, gradient stochastique, calcul distribué
- Fléau de la dimensionnalité...

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ \vdots & & & \\ x_{N1} & x_{N2} & \cdots & x_{Nd} \end{bmatrix}$$

- $d = 10^5, N = 10^4 \dots$
- Distribution des mots en fonction de leurs fréquences:



- Construire un système pour bien classer tous les documents proposés

- Il est souvent (toujours) possible de trouver des mots qui n'apparaissent que dans l'une des classes de documents...
- Il suffit de se baser dessus pour prendre une décision parfaite...
- Mais ces mots apparaissent ils dans les documents non vus jusqu'ici???

Idée:

Ajouter un terme sur la fonction coût (ou vraisemblance) pour pénaliser le nombre (ou le poids) des coefficients utilisés pour la décision

$$L_{\log} = \sum_{i=1}^N y_i \log(f(x_i)) + (1 - y_i) \log(1 - f(x_i)) - \lambda \|w\|_{\alpha}$$

$$C = \sum_{i=1}^N (f(x_i) - y_i)^2 + \lambda \|w\|_{\alpha}, \quad C = \sum_{i=1}^N (-y_i f(x_i))_+ + \lambda \|w\|_{\alpha}$$

Avec: $\|w\|_2 = \sum_j w_j^2$ ou $\|w\|_1 = \sum_j |w_j|$

- Etude de la mise à jour dans un algorithme de gradient
- On se focalise sur les coefficients *vraiment* important

Problème

Déséquilibre important entre les populations des classes dans l'ensemble d'entraînement

- Courant en pratique $\Rightarrow$ prédiction privilégiée de la classe dominante
- Ex: classification binaire avec 99% de $\ominus$: classifieur qui prédit toujours la classe dominante $\ominus \Rightarrow$ 99% accuracy
 - Utiliser d'autres métriques, ROC/AUC
- Comment améliorer l'entraînement ?
 - Ré-équilibrer le jeu de données: supprimer des données dans la classe majoritaire et/ou sur-échantillonner la classe minoritaire.
 - Changer la formulation de la fonction de coût pour pénaliser plus les erreurs dans la classe minoritaire. Avec $\Delta(f(x_i), y_i)$ la fonction de coût:

$$C = \sum_i \alpha_i \Delta(f(x_i), y_i), \quad \alpha_i = \begin{cases} 1 & \text{si } y_i \in \text{classe majoritaire} \\ B > 1 & \text{si } y_i \in \text{classe minoritaire} \end{cases}$$

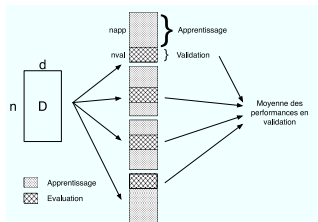
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■ Métriques d'évaluation

- Taux de reconnaissance $\frac{N_{correct}}{N_{tot}}$
- Précision (dans la classe c) $\frac{N_{correct}^c}{N_{predits}^c}$
- Rappel (dans la classe c) (=couverture) $\frac{N_{correct}^c}{N_{tot}^c}$
- F1 $\frac{(1+\beta^2)precision \cdot rappel}{\beta^2 precision + rappel}$
- ROC (faux pos VS vrai pos) / AUC

■ Procédures

- Apprentissage/test



- Validation croisée
- Leave-one-out

Regarder les poids des mots du classifieur:

annoying	37.2593
another	-8.458
any	3.391
anyone	-1.4651
anything	-15.5326
anyway	29.2124
apparently	12.5416
...	
attention	-1.2901
audience	1.7331
audiences	-3.7323
away	-14.9303
awful	30.8509

UE TAL :

Obligation de participer à une mini-compétition sur les 2 jeux de données

⇒ Buts:

- Traiter des données textuelles (!)
- Travail minimum d'optimisation des classifieurs
- Post-traitements & interactions (minimales) avec un système externe

- nltk
 - Corpus, ressources, listes de *stopwords*
 - quelques classifieurs (mais moins intéressant que sklearn)
- gensim
 - Très bonne implémentation (rapide)
 - Outils pour la sémantique statistique (cours suivants)
- sklearn
 - Boîte à outils de machine learning (SVM, Naive Bayes, regression logistique ...)
 - Evaluations diverses
 - Quelques outils pour le texte (simples mais pas très optimisés)

- Lancer des expériences à distance:
 - nohup (simple, mais perte du terminal)
 - Redirection, usage des logs
 - screen, tmux
- Connexion à distance = usage d'une passerelle
 - tunnel ssh
- Gestion des quotas
 - Travail sur le /tmp

⇒ Un rythme à trouver: fiabiliser le code en local, lancer les calculs lourds la nuit ou le week-end

- 1 Récupération/importation d'un corpus
 - Lecture de format XML
 - Template NLTK...
- 2 Optimisation d'un modèle.
 - Campagne d'expérience (d'abord grossière - codage, choix modèle...-, puis fine - régularisation...)
 - Assez long... Mais essentielle
 - **Le savoir-faire est ici**
- 3 Evaluation des performances (souvent en même temps que la phase d'optimisation)
 - Usage de la validation croisée
- 4 Apprentissage + packaging du modèle final
 - Définition des formats IO
 - Mode de fonctionnement : API, service web...
 - Documentation

- Montrer que vous êtes capables de réaliser une campagne d'expériences:
 - Courbes de performances
 - Analyse de ces courbes
- Montrer que vous êtes capable de valoriser un modèle
- Concrètement:
 - Mise au propre de votre code
 - Intégration des expériences dans une boucle (ou plusieurs)
 - Analyse qualitative du modèle final (tri des poids)
 - OPT: construction de nuages de mots...